

Milestone 2: Working Application

Studio Project

Milestone - Working application



Course Code: COSC3107|COSC3108

Weighting: 33.33% of your final mark for this studio project

Length: See Requirements Document

Task Type: Group of 2 students. Must not do each other's tasks; each student must complete tasks relating to their sub-task only and will be marked as such. Refer to section 3.3 of this page. Acceptability of group work depends on the group following the stated requirements.

Due date and time: Check this page or Canvas->Assignments for due dates that apply to you. A 10% late penalty applies for each 24 hour block late. After 5 days late, your submission might not be accepted. If special consideration is needed, you must apply via Canvas->'Assignment Extensions' as soon as possible. If granted, special consideration affects only the group member who applied and they must meet the university's [special consideration eligibility criteria](#)[Links to an external site.](#)  or have an [ELP](#) .

Feedback: Students must seek written clarifications via the course Canvas discussion board -> [Working Application Discussion Board](#).

Click or tap on the relevant sections to learn more

▼ 1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

- CL0 1: Apply foundational programming skills to design and develop small-to-medium-scale software programs using the Python programming language;
- CL0 2: Apply core principles of user-centred design to analyse and design user interfaces with good usability that consider human needs, capabilities, usability and user experience goals.
- CL0 4: Apply SQL as a programming language, directly and embedded in Python programs, to define database schemas, update database contents and extract data from databases for specific users' information needs;
- CL0 5: Demonstrate effective team participation, peer feedback, management and communication through relevant industry tools and practices, such as a version control system;
- CL0 6: Demonstrate adherence to appropriate standards and professional practice.

▼ 2. Overview of Assessment

The studio project involves developing a **data-driven website** to address a social challenge.

In this Milestone, you will:

1. Review the website specification document for the data-driven website to understand UCD and technical requirements the website needs to support.
2. Review the supplied database and metadata file to be used for your application.
3. Build your web application using HTML, CSS, and Python.
4. Link your web application to your database so users can complete the required tasks.
5. Demonstrate best practice group-based work with regular meetings, documentation, discussions & collaboration.
6. Submit a working application that is designed and built for a set of specific user groups.
7. Review and reflect on the UCD design elements of your implemented application.

▼ 3. Assessment Task

For this assessment, you will need to submit:

- Your team's web application by ensuring you have committed and pushed all your changes to your team's git repository before the due date.
- A brief report detailing UCD implementation and changes was submitted to Canvas before the due date.
- Project Log, showing your team tasks and progress during the implementation of Milestone 2.

3.1 Implementation of a Web Application

1. Review the requirements document for details on the levels and sub-tasks that need to be completed: [Studio Project Requirements Document](#).

2. Start on this assessment as soon as it is released by setting up / creating your Git repository and making initial commits from each student.
3. Review your wireframe designs using the feedback from your tutor/marker and use them as the basis for developing the pages in your web application.
4. Progressively develop your web application regularly by committing and pushing changes to your repository.
5. Complete the web pages for your level 1 & 2 tasks utilising HTML, CSS, Python and SQL
These must be fully functional and connected to your database (hard-coded data for Levels 1 & 2 is not sufficient.)
6. Complete a draft of your level 3 tasks utilising HTML, CSS, Python and SQL
Your level 3 pages can use hard-coded examples; for this submission, they do not need to have live data.
You will be required to make these pages have live data retrieved from the database for your final submission in Milestone 4.

3.2 Writing the Milestone 2 - UCD Implementation of Working Application report

Write a report that showcases your 1) UI design improvement, and 2) scenarios and storyboards. Create a Word document and share it in your Team's Channel Files.

Your report should include:

Part 1: UI Design improvements

For each subtask (A and B), show how you improved your design and why those changes matter.

- Select one page and describe the changes or improvements you made to its UI design.
- Provide annotated wireframes (from *Milestone 1*) and screenshots of your implemented web page, highlighting the changes or improvements.
- Explain how these modifications enhance the user experience or the completion of the subtask.
- Indicate whether each change relates to Nielsen's heuristics and/or a Navigation, Layout, or Interaction Design Pattern.

Part 2: Scenarios and Storyboards

For each subtask (A and B), demonstrate how your design supports the accomplishment of a real user's journey.

Select one Persona from *Milestone 1* and present the following:

- **Context Scenario:** Describe a realistic situation showing how your Persona interacts with the system to achieve a goal. Your scenario should address multiple levels of the subtask (Level 1 alone is not sufficient).
- **Key Path Scenario:** Based on your context scenario, outline the step-by-step flow your Persona follows, including specific pages, information, and interactions. Again, ensure it addresses multiple levels of the subtask (not only Level 1).
- **Storyboard:** Develop a storyboard that visually represents the key path scenario, aligning with the steps and interactions you described.

Your report must be no more than 7 pages, no less than 11 pt font and all the text and screenshots should be legible and readable by your assessor. Don't forget to add references if your images are generated by a GenAI tool, i.e., [RMIT Val](#) .

3.3 Teamwork Contribution

1. Use your Teams Channel for all text-based communication and video-based meetings. Aim to record the meetings you hold with your team. This will be used by your assessor as evidence of communication and discussions about the coordination and progress of your project.
2. Aim to schedule regular meetings (every 2-3 days) with your group. If you meet either online or in person, post the outcomes of the meeting within your Teams channel in a clear, organised and timely manner. Your assessor will look at these posts and the frequency of posts when marking your team participation.
3. Update the same "Project Log" document created in Milestone 1. Keep updating it with the new tasks and progress according to the Milestone 2 requirements. Ideally, every week you need to check on your progress and update the project log document. Your assessor will check on the updates and will assess the teamwork component based on your logged contributions.
4. Aim to make regular commits to your shared remote repository. If you have added your assessor as a contributor, they can look through your commit history and assess how active each member is. Thus, if you have not already, please add your assessor as a contributor of your GitHub repository.

Evidence of group work will only be accepted from your Teams Channel. Evidence of teamwork in other platforms, such as private messages, Discord, Instagram, or Snapchat, will not be considered. This is because staff require reliable access to evidence of group work on which to award grades.

3.4 Tips for Success

1. Building your HTML & CSS templates for all your pages.
2. Add hard-coded data examples to your templates to ensure your template is sufficient.
3. Incorporate your HTML, CSS & Python code so your web application is being dynamically built by your Python code.
4. Link your pages to your database so you can replace the hard-coded data with live data from your database.
5. Your completed application should respond to your user's queries and dynamically provide the data they requested. There should be no hard-coded data in this submission except for your level 3 pages.

▼ 4. Timeline

Note: You should have a team assessment plan.

Plan for the last commit and submission to be made well ahead of the deadline time.

After the final submission has been made, each team member should download the Canvas submission and test the latest version of your application.

Allow enough time so that if any issues are found, there is time to fix them.

To help you prepare for the assessment, please consider the following activities:

Activities

- Create static HTML files and CSS to use as templates for each of your pages; For now, you can hard-code the data that the pages need to show.
 - Start on the UCD implementation of Working Application Report
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- Review your requirements document and your wireframes.
 - Incorporate the static HTML templates and CSS into the web-app framework and test your navigation system
 - Complete your level 1 page to retrieve data from the DB using SQL
 - Draft your level 2 page (with hard-coded data for now).
 - Review and develop the UCD implementation of Working Application Report
 - Both students should follow the instructions in section 5 of this assignment page and make a draft submission of code and report.
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- Complete your level 2 page to retrieve data from the DB using SQL
 - Draft your level 3 page.
 - Review and provide feedback to your project partner on their pages.
 - Finalise the UCD implementation of Working Application Report
 - Both students should do a draft submission of code and report
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- Review your tasks and data requirements to fine tune your implementation
 - Check and test your application for any last minute issues.
 - Final submission of code and report

Activities

▼ 5. Assessment submission and feedback

You must follow all of the steps below for correct submission.

Preparation of Files:

1. Commit, pull, push, and merge all your final code to the main branch on Github and share the link to it in the below pdf.
2. Convert the Milestone 2 - UCD implementation of Working Application Report into a PDF and name it **studentID1_studentID2.pdf**
3. Update the final version of the Project Log file on your Teams Channel, showing the tasks and progress related to Milestone 2.

Important: It is the responsibility of both students to ensure that the above steps have been followed correctly.

▼ 6. Assessment Criteria

See attached rubric below.

▼ 7. Academic Integrity

Study Support Hub: Academic skills advisors and librarians will help you with researching your topic, finding information, improving your academic writing skills, academic integrity, effective learning strategies, statistics, and more!

Academic Integrity/plagiarism: You can achieve academic integrity by honestly submitting work that is your own. Presenting work that fails to acknowledge other people's work within yours can compromise academic integrity.

Special consideration: If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration.

Ready for Life and Work

Prepare students to work in software development projects.

For more information, please consult the course guide or contact your course coordinator.

Criteria	Ratings	Pts
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Criteria	Ratings						Pts
UCD 1. Annotated wireframes and final screenshots are directly linked to improvements and strong justifications of Nielsen's heuristics and/or design patterns. 2. Context scenario is detailed and aligned with the Personas and their needs and goals. It covers one or several levels of the subtasks. 3. Key path scenario is detailed and logically structured, clearly showing the webpages, information and interaction elements, and well aligned to the key path scenario. It covers one or several levels of the subtasks. 4. Storyboard effectively visualises the scenario with clear alignment to the key path.	4 pts High Distinction You have met all 4 sub-criterion with no significant improvements identified.	3.2 pts Distinction You have met 4 of the listed sub-criterion with no significant improvements identified. OR You have attempted all 4 of the listed sub-criterion, but areas of improvement have been identified.	2.8 pts Credit You have met 2 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 4 of the listed sub-criterion but areas of improvement have been identified for at least 2 of the sub-criterion.	2 pts Pass You have met 1 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 4 of the listed sub-criterion but areas of improvement have been identified for at least 1 of the sub-criterion	0.8 pts Partial Marks You have partially met 1 - 4 of the listed sub-criterion with significant improvements identified.	0 pts No marks There was no submission or the submission was not substantial enough to meet any of the sub-criterion.	4 pts

Criteria	Ratings						Pts
Python 1. The implementation enables the functionality for Level 1 and does not contain logical errors that produce an invalid response, such as incorrect data being returned. 2. The implementation enables the functionality for Level 2 and does not contain logical errors that produce an invalid response, such as incorrect data being returned. 3. The implementation demonstrates the intended functionality for Level 3.	6 pts High Distinction You have met all 5 sub-criterion with no significant improvements identified.	4.8 pts Distinction You have met 4 of the listed sub-criterion with no significant improvements identified. OR You have attempted all 5 of the listed sub-criterion but areas of improvement have been identified.	4.2 pts Credit You have met 3 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-criterion but areas of improvement have been identified for at least 2 of the sub-criterion.	3 pts Pass You have met 2 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-criterion but areas of improvement have been identified for at least 3 of the sub-criterion	1.2 pts Partial Marks You have met 1 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-criterion but areas of improvement have been identified for at least 4 of the sub-criterion	0 pts No marks There was no submission or the submission was not substantial enough to meet any of the sub-criterion.	6 pts
Database Implementation & SQL 1. SQL queries demonstrate retrieving raw data from a table and displaying it to the user. 2. SQL queries	5 pts High Distinction You have met all 5 sub-criterion with no significant improvements identified.	4 pts Distinction You have met 4 of the listed sub-criterion with no significant improvements identified. OR You have attempted all 5 of the listed sub-criterion	3.5 pts Credit You have met 3 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-	2.5 pts Pass You have met 2 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-	1 pts Partial Marks You have met 1 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-	0 pts No marks There was no submission or the submission was not substantial enough to meet any of the sub-criterion.	5 pts

Criteria	Ratings						Pts
demonstrate filtering data based on user input. 3. SQL queries demonstrate sorting data based on user input. 4. SQL queries demonstrate summarising data through the use of aggregation. 5. SQL queries demonstrate joining multiple tables together to retrieve a required dataset.		but areas of improvement have been identified.	criterion but areas of improvement have been identified for at least 2 of the sub-criterion.	criterion but areas of improvement have been identified for at least 3 of the sub-criterion	criterion but areas of improvement have been identified for at least 4 of the sub-criterion		
Teamwork 1. Weekly team meeting recordings (from week 12 onward) 2. Weekly team meeting minutes (from week 12 onward) 3. A detailed project plan that outlines key steps, timelines, and team member responsibilities for task. 4. Evidence has been provided that demonstrates the student took the initiative to seek and	5 pts High Distinction You have met all 5 sub-criterion with no significant improvements identified.	4 pts Distinction You have met 4 of the listed sub-criterion with no significant improvements identified. OR You have attempted all 5 of the listed sub-criterion but areas of improvement have been identified.	3.5 pts Credit You have met 3 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-criterion but areas of improvement have been identified for at least 2 of the sub-criterion.	2.5 pts Pass You have met 2 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-criterion but areas of improvement have been identified for at least 3 of the sub-criterion	1 pts Partial Marks You have met 1 of the listed sub-criterion with no significant improvements identified. OR You have attempted up to 5 of the listed sub-criterion but areas of improvement have been identified for at least 4 of the sub-criterion	0 pts No marks There was no submission or the submission was not substantial enough to meet any of the sub-criterion.	5 pts

Criteria	Ratings						Pts
provide feedback/assistance to their project partner.							
5. A group participation document for this submission which shows a table listing each group member and the percentage of work that they contributed to the overall submission (these must add up to 100%) Each group member must sign this document (e.g. a screenshot of acknowledgement in an online discussion tool)							
Total Points: 20							